AUTO SCALING FOR HIGH AVAILABILITY

AUTOSCALING WITH LOAD BALANCER

1.create a launch template

1. name----------webserver-LT
2. machine----------amazon-linux
3. security group---http/https/ssh
4. resource tag------Name-home
5. user-data

```
#!/bin/bash
sudo yum update -y
sudo yum install httpd -y
sudo systemctl start httpd.service
sudo echo "<h1> this is my webserver "$HOSTNAME" </h1>" >
/var/www/html/index.html
sudo systemctl restart httpd.service
sudo systemctl enable httpd.service
```

2.Create A Autoscaling Group

1. name -------ASG
2. template---webserver-LT
3. az and subnets----1a,1b,1c
4. capacity
   a. desired-------1
   b. minimum----1
   c. maximum----5
5. automatic scaling ------- target tracking scaling
   a. Metric type – average cpu utilization
   b. target value ----- 50%
6. create a autoscaling group

3.Create A Target Group

1.target type--------instance
2.target name------webserver-tg
3.path----------/
4.create a target group

4.Add Targets On Autoscaling Group

1. click on ASG
2. click on load balancing
3. add websever-tg

5.Create A Load Balancer

1. name----------ALB
2. default action----webserver-tg
3. create a load balancer

6.Add Load On Server

1. amazon-linux-extras install epel -y
2. yum install stress -y
3. stress --cpu 80 --timeout 330s

7.get load balancer dns and heat it on web

CUSTOM VPC WITH AUTOSCALING AND LOAD-BALANCER

1. **create a vpc (10.0.0.0/16)**
2. create a subnet
    1. public subnet-1 (10.0.0.0/20)---1a
    2. public subnet-2 (10.0.16.0/20)--1b
    3. private subnet-3 (10.0.32.0/20)--1a
    4. private-subnet-4 (10.0.48.0/20)--1b
3. create a internet getway and attach to a vpc
4. create a route table
    1. public-route
        a) subnet associate----public subnets
        b) route----internet getway
    2. private-route
        a) subnet associate----private subnets
5. **create a launch template**
    1. name -----> lt-home
    2. template-tags --------> Name-Home
    3. image--amazon-linux
    4. instance_type -----> t3.micro
    5. sg------> create a sg in my-vpc
    6. network_setting ----> enable-auto-assign-public_ip
    7. userdata

    #!/bin/bash

sudo yum update -y
sudo yum install httpd -y
sudo systemctl start httpd
sudo echo "<h1> this is my home webpage of "$HOSTNAME" </h1>" > /var/www/html/index.html
sudo systemctl restart httpd
sudo systemctl enable httpd

6. create a autoscaling group

1. name---> ASG
2. launch template---> lt-home
3. vpc----> my-vpc
4. Az & subnets ------> public-subnet-1 & public-subnet-2
5. desired---1, minimum--1, maximum—2
6. target-tracking-scaling ------> utilization (50%)
7. create a ASG

7. create a target group

1. name---> tg-home
2. type----> instance
3. vpc -----> my-vpc
4. create a target group

8. Assign target-group to ASG

1. click on ASG
2. click on integration
3. choose ----> application load balancer --> assign tg-home ----> update

9.create a load balancer

1. name---> ALB
2. vpc----> myvpc
3. mapping---> public subnet (1a) and public subnet (1b)
4. forward to ----tg-home
5. create a load balancer

10.copy dns and hit on the web

11. Test autoscaling

1.access the server

1. sudo yum install stress -y
2. stress --cpu 2 --timeout 300

APPLICATION INSIDE PRIVATE-SUBNET (VPC-AUTOSCALING-LOAD-BALANCER)

1. create a vpc (10.0.0.0/16)

1. create a subnet
    1. public subnet-1 (10.0.0.0/20)---ap-south-1a
    2. public subnet-2 (10.0.16.0/20)-- ap-south-1b
    3. private subnet-3 (10.0.32.0/20)-- ap-south-1a
    4. private-subnet-4 (10.0.48.0/20)-- ap-south-1b
2. create a internet getway and attach to a vpc
3. create a nat gateway in public-subnet-1
4. create a route table
    1. public-route
        a) subnet associate----public subnets (1 &2)
        b) route ----> internet-getway
    2. private-route
        a) subnet associate----private subnets (3&4)
        b) route ----> nat-gateway

2. create a launch template

1. name -----> lt-home
2. template-tags --------> Name-Home
3. image-----> amazon-linux
4. instance_type -----> t3.micro
5. sg------> create a sg in my-vpc
6. network_setting ----> disable-auto-assign-public_ip
7. userdata

```
#!/bin/bash
yum install httpd -y
systemctl start httpd
systemctl enable httpd
echo "<h1>WELCOME TO AUTOSCALING PROJECT</h1>" >
/var/www/html/index.html
echo "<h2>Hostname: $(hostname)</h2>" >> /var/www/html/index.html
```

3. create a autoscaling group

1. name---> ASG
2. launch template---> lt-home
3. vpc----> my-vpc
4. Az & subnets ------> private-subnet-3 & private-subnet-4
5. desired---1, minimum--1, maximum--2
6. target-tracking-scaling ------> utilization (50%)

7. create a ASG

4. create a target group

1. name---> tg-home
2. type----> instance
3. vpc -----> my-vpc
4. create a target group

5. Assign target-group to ASG

1. click on ASG
2. click on integration
3. choose ----> application load balancer --> assign tg-home ----> update

6.create a load balancer

1. name---> ALB
2. vpc------> my-vpc
3. mapping------> public-subnet-1 (1a) and public-subnet-2 (1b)
4. forward to ---> tg-home
5. create a load balancer

7.copy dns and hit on the web

http://ALB-DNS

8. Test autoscaling

1. access the private-server using bastion-host-server
    1. sudo yum install stress -y
    2. stress --cpu 2 --timeout 300

QUESTIONS HOW ALB ACCESS PRIVATE-SERVERS

1. WHY APPLICATION SERVERS ARE IN PRIVATE SUBNET?

- Better Security
- No direct internet access
- Protect from hackers and attacks
- Real production architecture

2. CAN INTERNET ACCESS PRIVATE EC2 DIRECTLY?

NO

Because:

- Private EC2 has no public IP
- Private subnet has no route to Internet Gateway

3. THEN HOW DOES WEBSITE WORK?

User → ALB → Target Group → Private EC2

- ALB is public
- EC2 is private
- ALB forwards traffic internally

4. HOW ALB COMMUNICATES WITH PRIVATE EC2?

- Because both are inside same VPC.
- Communication happens using: Private IP addresses

5. WHAT IS TARGET GROUP?

- Target Group connects:
  ALB ↔ EC2 Instances
- ALB sends traffic to Target Group.
- Target Group sends traffic to EC2.

6. WHY NAT GATEWAY IS USED?

Private EC2 needs internet for:

- yum update
- package installation
- API calls

Flow:

Private EC2 → NAT Gateway → Internet

ARCHITECTURE

Internet
↓
ALB (Public Subnet)
↓
Private EC2 (Private Subnet)
↓
Private Database
